

Case - Minimization with an Equality Constraint

Tableau Simplex (Big-M) solution report · generated 2026-09-27 13:53

Demonstrates 'multiply the objective by -1 and maximize' (lecture slide 9) together with $\geq$ and $=$ constraints (two artificial variables).

Model

Minimize $Z = 2X_1 + 3X_2$
subject to
[C1] $0.5X_1 + 0.25X_2 \leq 4$
[C2] $X_1 + 3X_2 \geq 20$
[C3] $X_1 + X_2 = 10$
 $X_1, X_2 \geq 0$

RESULT

OPTIMAL

$Z^* = 25$

$X_1 = 5$

$X_2 = 5$

iterations: 2 · rule: dantzig

Findings

- Optimal: row 0 has no negative coefficients. $Z^* = 25$ (converted back: $Z = -Z'$)

Contents: standard form · geometry · every tableau with entering/leaving analysis · decision diagram · summary dashboard · independent verification

1. Converting to the standard LP form

1. The objective is minimisation → multiply by -1 and maximise $Z' = -Z$ (lecture slide 9). The final answer is converted back with $Z = -Z'$.
2. C1 (Type I, $\leq$): add slack $X3 \geq 0$ (unused resource, cost 0) → $X3$ is the starting basic variable of this row.
3. C2 (Type II, $\geq$): subtract surplus $X4 \geq 0$ (cost 0). Its column is -1 , so it cannot start in the basis → an artificial variable is needed.
4. C3 (=): already an equation, but it has no obvious basic variable → an artificial variable is needed.
5. C2: add artificial $X5 \geq 0$ with objective coefficient $-M$ (huge penalty) → it provides an immediately recognisable starting basic variable.
6. C3: add artificial $X6 \geq 0$ with objective coefficient $-M$ (huge penalty) → it provides an immediately recognisable starting basic variable.

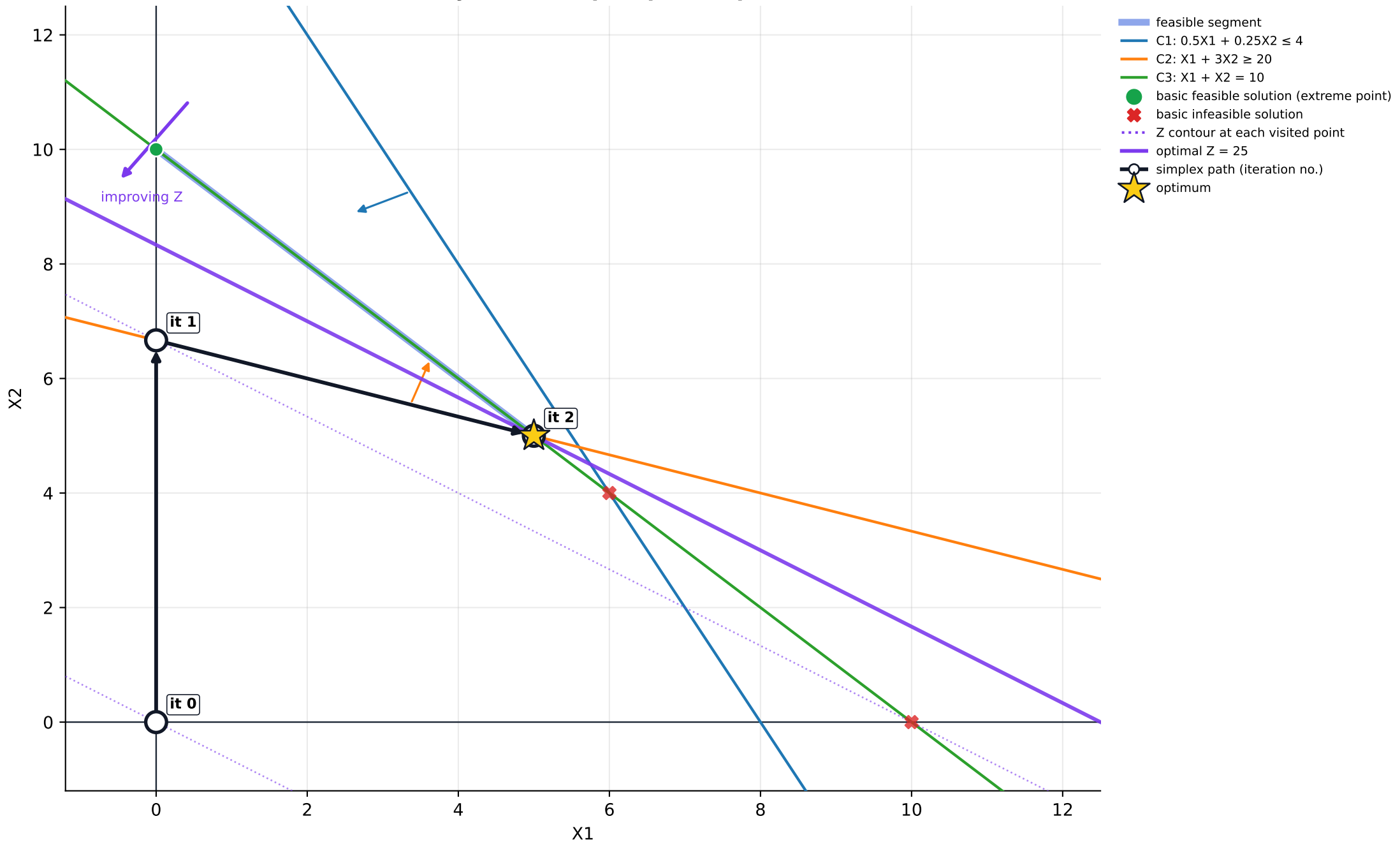
Maximize $Z' = -2X1 - 3X2 - MX5 - MX6$
 subject to
 $0.5X1 + 0.25X2 + X3 = 4$
 $X1 + 3X2 - X4 + X5 = 20$
 $X1 + X2 + X6 = 10$
 $X1, X2, X3, X4, X5, X6 \geq 0$

Standard form - coefficient matrix

	X1	X2	X3	X4	X5	X6		RHS
max Z'	-2	-3	0	0	-M	-M		
C1	0.5	0.25	1	0	0	0	=	4
C2	1	3	0	-1	1	0	=	20
C3	1	1	0	0	0	1	=	10

2. Geometric view: basic solutions, extreme points and the simplex path

Geometry of the simplex path - optimal



3. All basic solutions (m variables basic, the rest 0). Green = feasible = extreme point; red = infeasible; ★ = optimal

Basic variables	X1	X2	Feasible?	Z	X3	X4	Optimal?
X1, X2, X3	5	5	yes	25	0.25	0	★
X1, X2, X4	6	4	no	24	0	-2	
X1, X3, X4	10	0	no	20	-1	-10	
X2, X3, X4	0	10	yes	30	1.5	10	

Initial tableau

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS
Z	0	1	2	3	0	0	M	M	0
X3	1	0	0.5	0.25	1	0	0	0	4
X5	2	0	1	3	0	-1	1	0	20
X6	3	0	1	1	0	0	0	1	10

Initial tableau. Basic variables: $X3 = 4$, $X5 = 20$, $X6 = 10$. Non-basic (set to 0): $X1$, $X2$, $X4$. The values of the basic variables are read directly from the RHS column.

Initial tableau after Big-M preparation

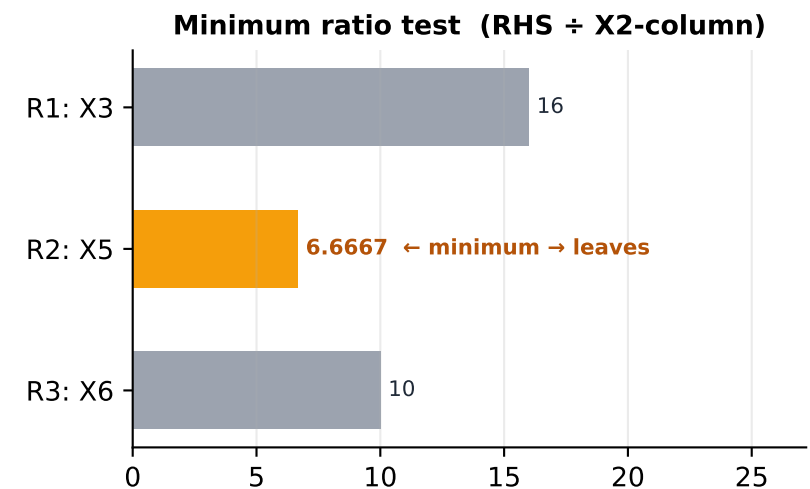
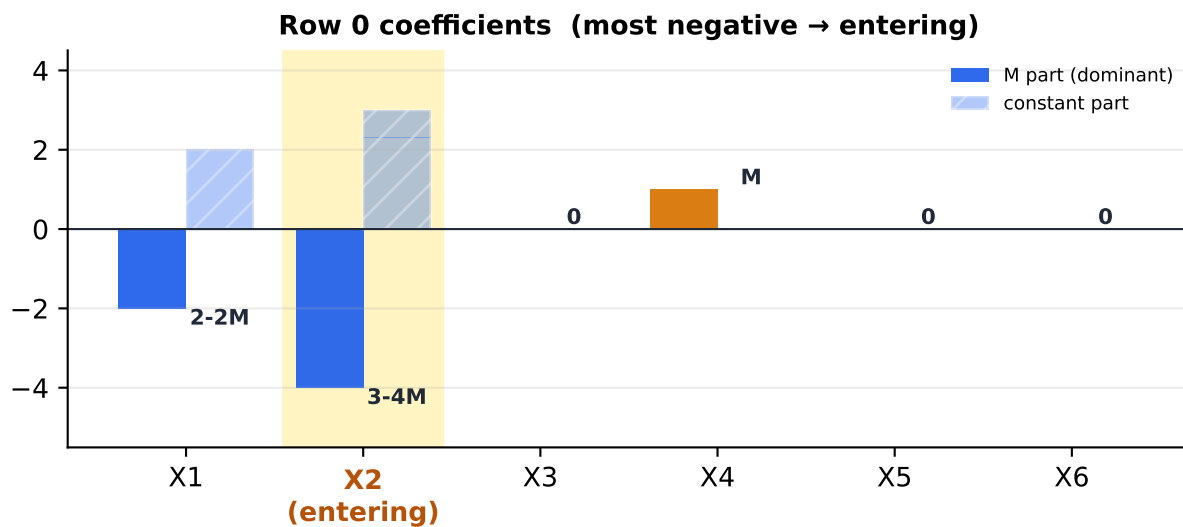
Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS
Z	0	1	2-2M	3-4M	0	M	0	0	-30M
X3	1	0	0.5	0.25	1	0	0	0	4
X5	2	0	1	3	0	-1	1	0	20
X6	3	0	1	1	0	0	0	1	10

Big-M preparation. the artificial basic variable(s) had coefficient M in row 0. Subtracting M × their rows makes those coefficients zero, so row 0 is ready for the optimality test.

- Subtract M × R2 from R0 (make the coefficient of basic X5 zero)
- Subtract M × R3 from R0 (make the coefficient of basic X6 zero)

Iteration 1: entering & leaving variables

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS	Ratio
Z	0	1	2-2M	3-4M	0	M	0	0	-30M	
X3	1	0	0.5	0.25	1	0	0	0	4	16
X5	2	0	1	3	0	-1	1	0	20	6.6667
X6	3	0	1	1	0	0	0	1	10	10



Iteration 1 - selecting variables.

- Is the current solution optimal? No - row 0 has at least one negative coefficient.
- Entering basic variable: X2 - it has the largest (in absolute value) negative coefficient in row 0 (3-4M).
- Minimum ratio test (RHS ÷ positive entries of the X2-column): R1: 16, R2: 6.6667, R3: 10.
- Leaving basic variable: X5 (row 2, smallest ratio). Pivot number = 3.

Iteration 1: pivoting (Gauss-Jordan row operations)

(a) pivot row scaled: Divide R2 by the pivot number 3

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS	Ratio
Z	0	1	2-2M	3-4M	0	M	0	0	-30M	
X3	1	0	0.5	0.25	1	0	0	0	4	16
X5	2	0	0.3333	1	0	-0.3333	0.3333	0	6.6667	6.6667
X6	3	0	1	1	0	0	0	1	10	10

(b) X2 replaced X5 in the basis. Z = -20-3.3333M

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS
Z	0	1	1-0.6667M	0	0	1-0.3333M	-1+1.3333M	0	-20-3.3333M
X3	1	0	0.4167	0	1	0.0833	-0.0833	0	2.3333
X2	2	0	0.3333	1	0	-0.3333	0.3333	0	6.6667
X6	3	0	0.6667	0	0	0.3333	-0.3333	1	3.3333

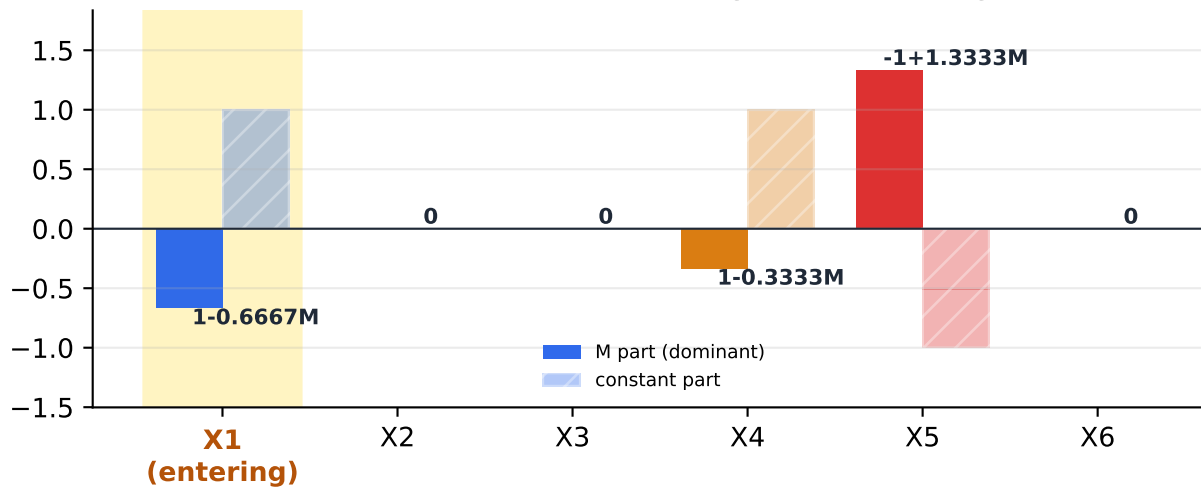
Row operations:

- Add $(-3+4M) \times R2$ to R0
- Subtract $0.25 \times R2$ from R1
- Subtract $1 \times R2$ from R3

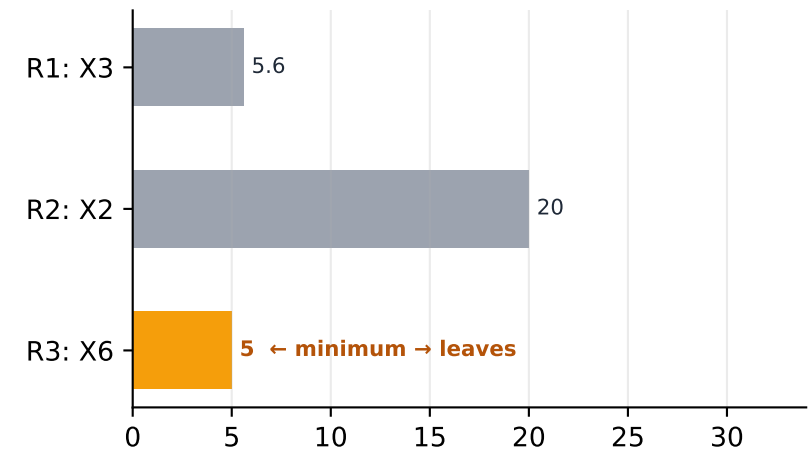
Iteration 2: entering & leaving variables

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS	Ratio
Z	0	1	1-0.6667M	0	0	1-0.3333M	-1+1.3333M	0	-20-3.3333M	
X3	1	0	0.4167	0	1	0.0833	-0.0833	0	2.3333	5.6
X2	2	0	0.3333	1	0	-0.3333	0.3333	0	6.6667	20
X6	3	0	0.6667	0	0	0.3333	-0.3333	1	3.3333	5

Row 0 coefficients (most negative → entering)



Minimum ratio test (RHS ÷ X1-column)



Iteration 2 - selecting variables.

- Is the current solution optimal? No - row 0 has at least one negative coefficient.
- Entering basic variable: X1 - it has the largest (in absolute value) negative coefficient in row 0 (1-0.6667M).
- Minimum ratio test (RHS ÷ positive entries of the X1-column): R1: 5.6, R2: 20, R3: 5.
- Leaving basic variable: X6 (row 3, smallest ratio). Pivot number = 0.6667.

Iteration 2: pivoting (Gauss-Jordan row operations)

(a) pivot row scaled: Divide R3 by the pivot number 0.6667

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS	Ratio
Z	0	1	$1-0.6667M$	0	0	$1-0.3333M$	$-1+1.3333M$	0	$-20-3.3333M$	
X3	1	0	0.4167	0	1	0.0833	-0.0833	0	2.3333	5.6
X2	2	0	0.3333	1	0	-0.3333	0.3333	0	6.6667	20
X6	3	0	1	0	0	0.5	-0.5	1.5	5	5

(b) X1 replaced X6 in the basis. Z = -25

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS
Z	0	1	0	0	0	0.5	$-0.5+M$	$-1.5+M$	-25
X3	1	0	0	0	1	-0.125	0.125	-0.625	0.25
X2	2	0	0	1	0	-0.5	0.5	-0.5	5
X1	3	0	1	0	0	0.5	-0.5	1.5	5

Row operations:

- Add $(-1+0.6667M) \times R3$ to R0
- Subtract $0.4167 \times R3$ from R1
- Subtract $0.3333 \times R3$ from R2

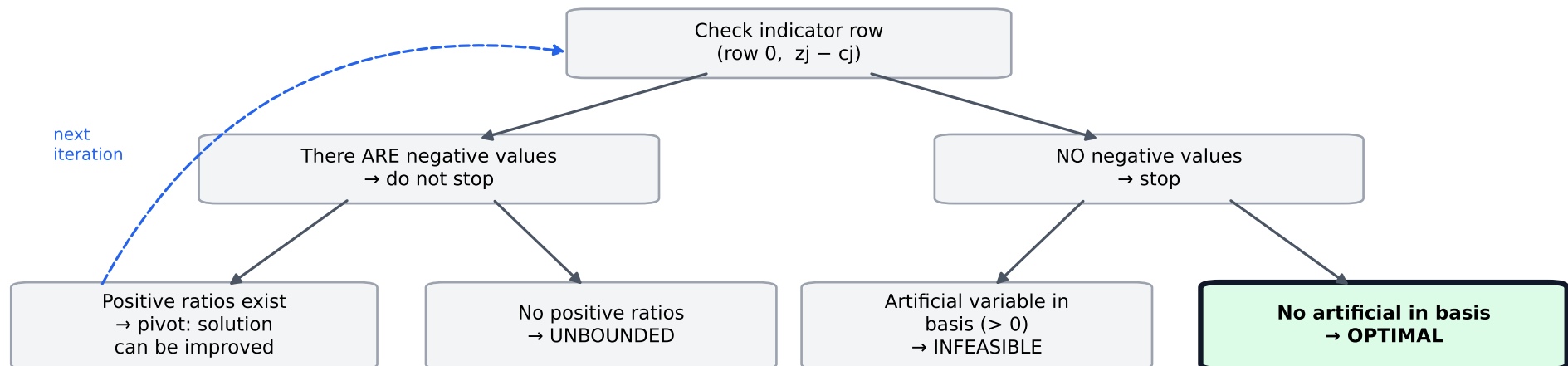
Final tableau - OPTIMAL

Basic	Eq.	Z	X1	X2	X3	X4	X5	X6	RHS
Z	0	1	0	0	0	0.5	-0.5+M	-1.5+M	-25
X3	1	0	0	0	1	-0.125	0.125	-0.625	0.25
X2	2	0	0	1	0	-0.5	0.5	-0.5	5
X1	3	0	1	0	0	0.5	-0.5	1.5	5

Final check.

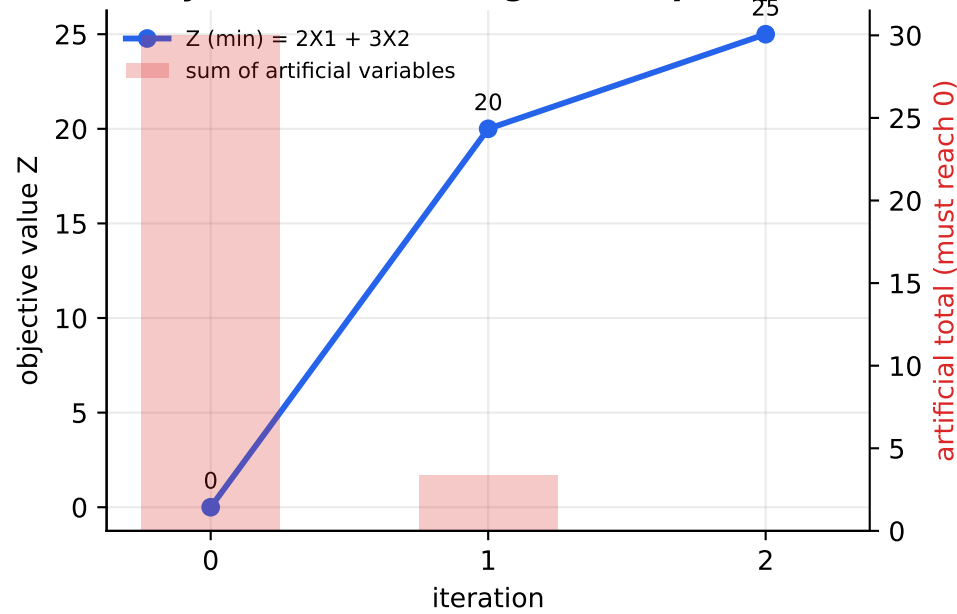
- Optimal: row 0 has no negative coefficients. $Z^* = 25$ (converted back: $Z = -Z'$)

Simplex decision diagram - outcome: OPTIMAL

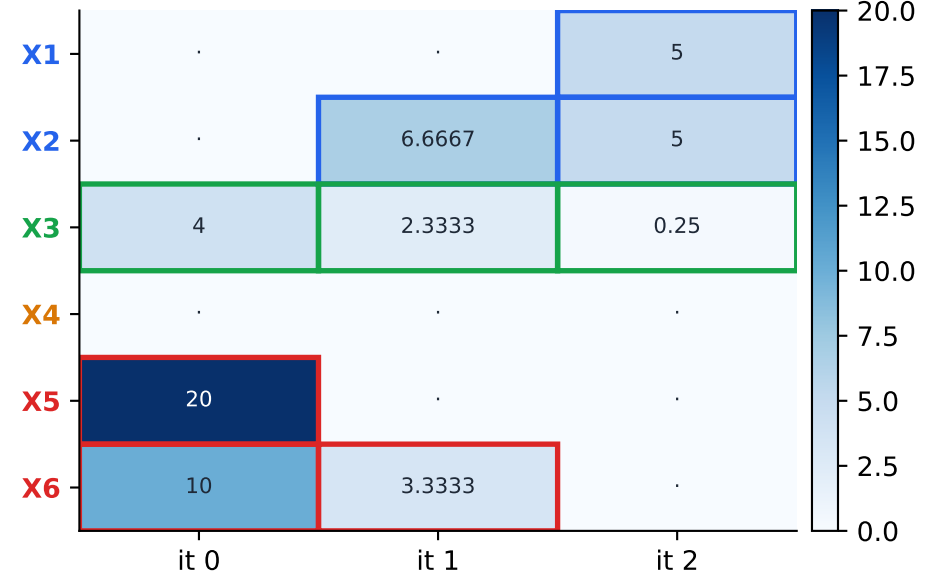


Case - Minimization with an Equality Constraint - OPTIMAL, $Z^* = 25$

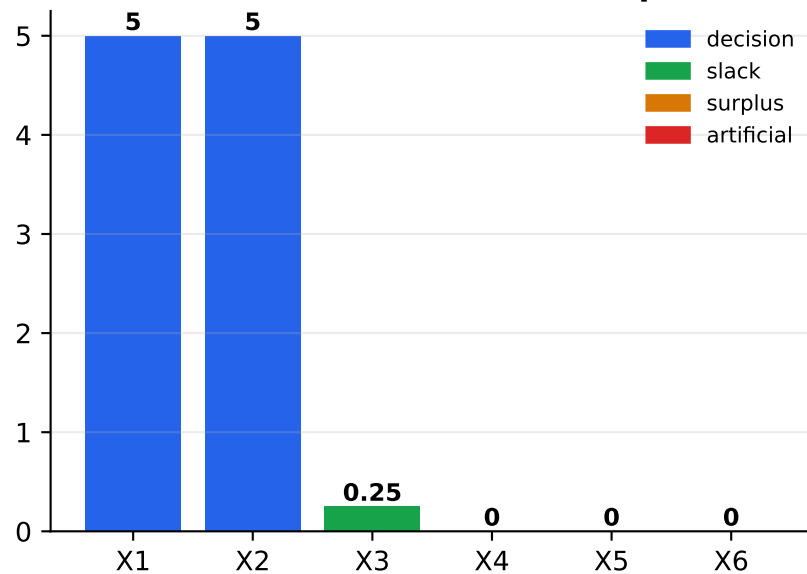
Objective value along the simplex path



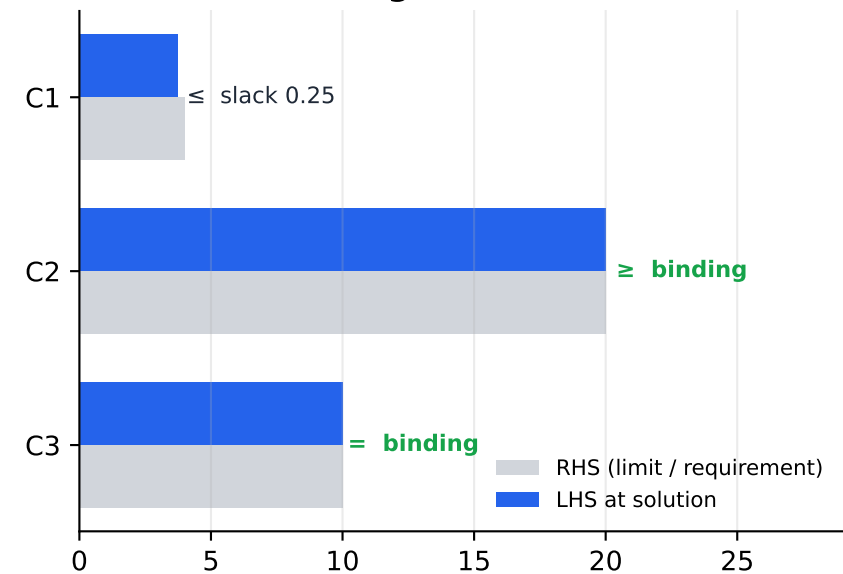
Basis evolution (outlined = basic variable)



Final values of all variables (optimal)



Constraint usage at the final solution



Independent verification with SciPy (HiGHS)

quantity	our tableau simplex	SciPy HiGHS
status	optimal	optimal
Z	25	25
X1	5	5
X2	5	5

✓ **Results agree**